

Untitled Document

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Definition. We say that an integer n is even if there exists an integer m such that $n = 2m$.

Example. 6 is even. In the definition, $n = 6$. We need to answer the question: “is there an integer m such that $6 = 2m$? Sure: if $m = 3$, then $n = 2m$.

Is 7 even? Can we solve the equation $7 = 2m$? Yes.. $m = 7/2$. Problem: this is not an integer, so 7 is not even.

Is -8 even? Sure: $-8 = 2(-4)$ ($m = -4$).

Is 0 even? Yes. In the definition, $n = 0$, and we need to solve the equation $n = 2m$, i.e., the equation $0 = 2m$. The choice $m = 0$ solves this equation. Thus, 0 is even.

Example of LaTeX. $n^2 + m_1 + \alpha\beta\gamma\Gamma$.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Lemma. Suppose that n and m are even integers. Then $n + m$ is even.

Assume that n and m are even integers. Since n is even, there exists an integer a such that $n = 2a$. Since m is even, there exists an integer b such that $m = 2b$. Thus, $n + m = 2a + 2b$. By the distributive law, $n + m = 2(a + b)$. If we let $c = a + b$, then $n + m = 2c$, and c is an integer (because the sum of 2 integers is an integer). We conclude that $n + m$ is even.

Other symbols. For sets, we use bold. The set of real numbers is **R**. The complex number are **C**. The rationals are **Q**. The integers are **Z**; $3 \in \mathbf{Z}$, $\pi \in \mathbf{R}$; $\pi \notin \mathbf{Z}$; $\mathbf{Z} \neq \mathbf{R}$. There exists is $\exists$; $\nexists$; for all $\forall$; $\mathbf{Z} \subset \mathbf{R}$